

NAME: _____

PERIOD: _____

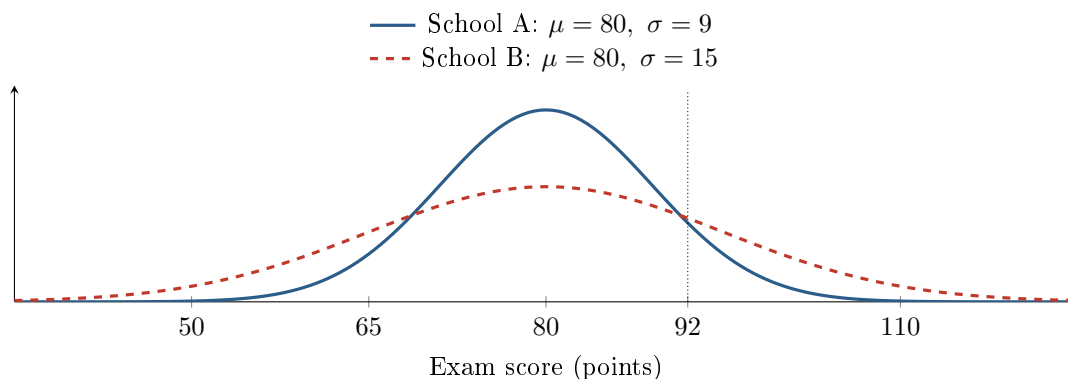
Comparing a Score of 92

AP Statistics · Unit 2 · Topic 2.11

[STAR] THE PROBLEM

Suppose exam scores at School A and School B are each approximately normal. Both distributions have mean 80 points, but School A has standard deviation 9 points and School B has standard deviation 15 points.

A counselor says, “A score of 92 is at about the 90th percentile at both schools because both schools have the same mean.” A data coach says, “The different spreads could put 92 at different percentiles.”



FIRST TAKE — one sentence before you work the parts

Would a 92 stand out equally at both schools? Give one reason from the picture.

[BOOK] STANDARDIZE BEFORE COMPARING (keep on the page)

For a normal random variable, $z = \frac{x - \mu}{\sigma}$ tells how many standard deviations x is from its mean. The percentile of x is the percent of values at or below x . Use a normal table or calculator to find the area to the left of the z -score; multiply that area by 100. A larger percentile means a higher standing relative to that school's scores. Printed normal table: $P(Z \leq 0.8000) = 0.788145$ and $P(Z \leq 1.3333) = 0.908789$. These left areas supply the lookup needed above.

Work (a)–(c).

DOK 1 (a) Identify which school has the more variable score distribution and state how the corresponding normal curve shows that difference.

DOK 2 (b) For a score of 92, calculate the z -score and percentile at each school. Show the standardization and report each percentile to the nearest tenth of a percent.

DOK 3 (c) In 3–4 sentences, decide whose claim is better supported. Use both percentiles from (b) and explain how the different spreads change what a score of 92 means at each school.

Frame: At School A, $z =$ _____ and the percentile is _____; at School B, $z =$ _____ and the percentile is _____.

Frame: The counselor's claim is _____ because the same 12-point gap is _____; therefore a 92 has the higher relative standing at _____.

Turn this sheet in whenever you finish — bonus credit. Part (c) is scored E / P / I.